

(ACT, EC, EE, VLSI)

Max. Marks: 100

20 Marks

- 1-

10. If $L\{f(t)\} = F(s)$, then $L\left\{\int_0^t f(t) dt\right\} =$ _____

A) $F'(s)$
 C) $\frac{F(s)}{s}$

B) $s F(s)$
 D) $\int_s^\infty F(s) ds$

11. Laplace transform of periodic function $f(t)$ with period T is _____

A) $\frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt$

B) $\frac{1}{1 + e^{-sT}} \int_0^T e^{-st} f(t) dt$

C) $\frac{1}{1 - e^{-sT}} \int_0^t e^{-st} f(t) dt$

D) $\frac{1}{1 - e^{-sT}} \int_0^\infty e^{-st} f(t) dt$

12. What is $L\{u(t - a)\} =$

A) e^{-as}
 C) $\frac{e^{as}}{s}$

B) se^{-as}
 D) $\frac{e^{-as}}{s}$

13. $L^{-1}\left\{\frac{1}{s^n}\right\} =$ _____

A) t^{n-1}

B) $\frac{t^{n-1}}{(n-1)!}$
 D) $\frac{t^{n+1}}{(n+1)!}$

C) t^{n+1}

14. Value of $L^{-1}\left\{-\frac{d}{ds}(F(s))\right\}$

A) $t f(t)$
 C) $\frac{f(t)}{t}$

B) $f(t)$
 D) $-f(t)$

15. Find $L^{-1}\left\{\frac{e^{-s}}{s}\right\}$

A) $u(t-1)(t-1)$
 C) $u(t-1)$

B) $u(t+1)(t+1)$
 D) $u(t+1)$

16. Value of $L\{f'(t)\} =$

A) $s L\{f(t)\} - f(0)$
 C) $s L\{f(t)\} - f'(0)$

B) $s L\{f(t)\} + f(0)$
 D) $s L\{f(t)\} + f'(0)$

17. The PDE obtained by eliminating arbitrary constants from $z = ax^2 + by^2$ is

A) $z = px + qy$
 C) $z = px - qy$

B) $2z = px + qy$
 D) $2z = px - qy$

18. The equation of the one-dimensional wave equation is

A) $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$

B) $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$

C) $\frac{\partial^2 u}{\partial x^2} = c^2 \frac{\partial^2 u}{\partial t^2}$

D) $\frac{\partial u}{\partial x} = c^2 \frac{\partial^2 u}{\partial t^2}$

19. The general solution of $Pp + Qq = R$, where P, Q, R are functions of x, y, z , is

A) $F(u, v) = 1$, Where u, v are functions of x, y, z .
 C) $F(u, v) = 0$, Where u, v are functions of x, y, z .

B) $F(u + v) = 0$, Where u, v are functions of x, y, z .
 D) $F(u - v) = 0$, Where u, v are functions of x, y, z .

MA1003-1

Summer SEE July 2024

20. The auxiliary equation of $Pp + Qq = R$ is of the form

A) $\frac{dx}{p} = \frac{dy}{q} = \frac{dz}{R}$
 C) $\frac{dx}{p} + \frac{dy}{q} + \frac{dz}{R}$

B) $\frac{dx}{P} + \frac{dy}{Q} + \frac{dz}{R}$
 D) $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

PART - B: DESCRIPTIVE ANSWER QUESTIONS**Unit – I**

	Marks	BT*	CO*	PO*
1. a) Solve $\frac{dy}{dx} + y \tan x = y^3 \sec x$.	6	L*3	1	2
b) If disintegration at any instant is proportional to the amount of material present and if 30% of the radio active substance disappeared in 10 days, find how long will it take for 90% of it to disappear.	6	L2	1	2
c) Solve the differential equation $(D^3 - 2D^2 + 4D - 8)y = 0$.	4	L1	2	1
2. a) Using the method of variation of parameters solve the differential equation $(D^2 - 4D + 4)y = \frac{e^{2x}}{x}$.	6	L2	2	2
b) Solve the differential equation $(D^3 - D)y = (2x + 1) + 4 \cos x$.	6	L3	2	2
c) Find the singular solution of the differential equation $p = \log(px - y)$.	4	L1	1	1
3. a) Solve $x^2 p^2 + x y p - 6y^2 = 0$.	6	L2	1	1
b) Solve the Cauchy's homogeneous linear equation $x^2 \frac{d^2 y}{dx^2} - 2y = x^2 + \frac{1}{x}$.	6	L3	2	2
c) Obtain the orthogonal trajectories of the family of curves $r^n = a \sin n\theta$.	4	L1	1	2

Unit – II

4. a) Determine $L\{t^2 e^{-3t} \sin 2t\}$.	6	L2	3	2
b) If $f(t)$ is a periodic function with period T , then prove that $L\{f(t)\} = \frac{1}{1-e^{-sT}} \int_0^T e^{-st} f(t) dt$.	6	L1	3	1
c) Solve $L^{-1}\left\{\log\left(\frac{s+a}{s+b}\right)\right\}$.	4	L2	4	1
5. a) Find $L^{-1}\left\{\frac{3}{s} - \frac{4e^{-s}}{s^2} + 4\frac{e^{-3s}}{s^2}\right\}$.	6	L3	4	1
b) Using convolution theorem of Laplace inverse transforms, find $L^{-1}\left\{\frac{1}{s^2(s-1)}\right\}$.	6	L2	4	2
c) Find $L\{e^{-3t} \sin 5t \sin 3t\}$.	4	L1	3	1
6. a) Rewrite the following using unit step function and find its Laplace transforms.				
$f(t) = \begin{cases} t-1, & 0 \leq t < 2 \\ 3-t, & 2 \leq t < 3 \\ 0, & t \geq 3 \end{cases}$	6	L2	3	2
b) Solve $x''(t) + 4x'(t) + 4x(t) = 4e^{-2t}$, given $x(0) = -1$, $x'(0) = 4$.	6	L3	4	2
c) Find $L\left\{\int_0^t \frac{e^t \sin t}{t} dt\right\}$.	4	L1	3	1

Unit – III

7. a) Solve the following partial differential equation by direct integration.				
$\frac{\partial^3 z}{\partial x^2 \partial y} + 18xy^2 + \sin(2x - y) = 0$	6	L3	5	2
b) Derive one dimensional wave equation				
$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$	6	L2	5	1
c) Form the partial differential equation by eliminating arbitrary function from $x + y + z = f(x^2 + y^2 + z^2)$.	4	L1	5	1
8. a) Solve the following PDE by the method of separation of variables				
$y^3 \frac{\partial z}{\partial x} + x^2 \frac{\partial z}{\partial y} = 0$	6	L2	5	2
b) Solve the following linear partial differential equation by Lagrange's method				
$(z - y) \frac{\partial z}{\partial x} + (x - z) \frac{\partial z}{\partial y} = y - x$	6	L3	5	1
c) Solve the following non-linear PDE $p(1 + q) = qz$.	4	L1	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
